

Notes on Model Validation in Bayesian Inverse Problems

With Application to Disorder Inference in Majorana Nanowires

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1 Setting and Motivation

1.1 The general Bayesian inverse problem

We consider the standard framework for Bayesian inversion. Let $\theta \in \Theta$ be an unknown parameter (possibly infinite-dimensional), $\mathcal{G} : \Theta \rightarrow \mathbb{R}^d$ a forward map, and $\mathbf{y} \in \mathbb{R}^d$ observed data related to θ by the measurement model

$$\mathbf{y} = \mathcal{G}(\theta) + \eta, \quad \eta \sim p_\eta, \quad (1)$$

where η is observation noise with known (or partially known) distribution p_η . In the Gaussian case, $\eta \sim N(0, \Sigma)$ with known covariance $\Sigma \in \mathbb{R}^{d \times d}$, and the likelihood is

$$L(\mathbf{y} \mid \theta) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2} \|\Sigma^{-1/2}(\mathbf{y} - \mathcal{G}(\theta))\|^2\right). \quad (2)$$

Given a prior distribution $\pi_0(\theta)$, the posterior is

$$\pi(\theta \mid \mathbf{y}) = \frac{L(\mathbf{y} \mid \theta) \pi_0(\theta)}{Z}, \quad Z := \int_{\Theta} L(\mathbf{y} \mid \theta) \pi_0(\theta) d\theta, \quad (3)$$

where Z is the **evidence** (marginal likelihood). In practice, $\pi(\theta \mid \mathbf{y})$ is explored via MCMC, which produces samples $\theta^{(1)}, \dots, \theta^{(K)} \sim \pi(\theta \mid \mathbf{y})$ but does not directly compute Z .

1.2 The model criticism question

The posterior (3) is the optimal answer to the question: “Given that the measurement model (1) is correct, what can we learn about θ ?” But it does not answer the prior question: *is the model correct?*

Model misspecification can take several forms:

- (i) The forward map $\mathcal{G}$ does not capture the true physics (e.g., the physical model is missing degrees of freedom, or uses an incorrect constitutive law).
- (ii) The noise model p_η is wrong (e.g., the noise is non-Gaussian, or the covariance Σ is misspecified).
- (iii) The prior π_0 is too restrictive (e.g., the true parameter lies outside the support of π_0 , or the smoothness assumptions are wrong).
- (iv) The parameterization of θ is insufficient (e.g., the unknown is really a function but is being approximated by too few basis coefficients).

This document describes four complementary approaches to detecting and diagnosing these failures within the Bayesian framework: posterior predictive checks (§2), Bayesian model comparison via the evidence (§3), hierarchical inference of hyperparameters (§4), and residual analysis (§5).

2 Posterior Predictive Checks

2.1 The posterior predictive distribution

Given posterior samples $\theta^{(k)} \sim \pi(\theta \mid \mathbf{y})$, the **posterior predictive distribution** is the distribution of hypothetical future data $\mathbf{y}^{\text{rep}}$ generated under the fitted model:

$$p(\mathbf{y}^{\text{rep}} \mid \mathbf{y}) = \int L(\mathbf{y}^{\text{rep}} \mid \theta) \pi(\theta \mid \mathbf{y}) d\theta. \quad (4)$$

In practice this integral is evaluated by Monte Carlo: for each posterior sample $\theta^{(k)}$, draw $\mathbf{y}^{(k)} \sim L(\cdot \mid \theta^{(k)})$, i.e.,

$$\theta^{(k)} \sim \pi(\theta \mid \mathbf{y}), \quad \mathbf{y}^{(k)} = \mathcal{G}(\theta^{(k)}) + \eta^{(k)}, \quad \eta^{(k)} \sim p_\eta \text{ (fresh, independent noise)}. \quad (5)$$

The collection $\{\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(K)}\}$ is a sample from (4).

2.2 Summary statistics and the posterior predictive p-value

The key idea: if the model is correct, the observed data $\mathbf{y}$ should be a “typical” draw from the posterior predictive distribution (4). To make this precise, choose a scalar **summary statistic** (also called a **test quantity**) $T : \mathbb{R}^d \rightarrow \mathbb{R}$ and compare $T(\mathbf{y})$ to the distribution of $T(\mathbf{y}^{(k)})$.

Step 1. Generate K replicated datasets $\mathbf{y}^{(1)}, \dots, \mathbf{y}^{(K)}$ via (5). (You already have the posterior samples from MCMC; the only additional cost is evaluating $\mathcal{G}(\theta^{(k)})$, which you have already done during the MCMC run, plus drawing fresh noise $\eta^{(k)}$.)

Step 2. Compute the histogram of $\{T(\mathbf{y}^{(1)}), \dots, T(\mathbf{y}^{(K)})\}$.

Step 3. Compute $T(\mathbf{y})$ for the actual observed data.

Step 4. Report the **posterior predictive p-value**:

$$p_B := \frac{1}{K} \sum_{k=1}^K \mathbf{1}[T(\mathbf{y}^{(k)}) \geq T(\mathbf{y})]. \quad (6)$$

Interpretation. If p_B is close to 0 or 1, the observed data is in the tail of the predictive distribution for that statistic, which is evidence that the model cannot reproduce the feature captured by T . Values in $[0.1, 0.9]$ indicate no evidence of misspecification (for that particular T).

2.3 How to choose the summary statistic T

The choice of T is the art of posterior predictive checking. There is no single “right” choice; different statistics probe different aspects of the model. The general principles are:

- (a) **Omnibus statistics** test overall fit. The most common is the chi-squared discrepancy:

$$T_{\chi^2}(\mathbf{y}, \theta) = \|\Sigma^{-1/2}(\mathbf{y} - \mathcal{G}(\theta))\|^2. \quad (7)$$

Note that this T depends on both $\mathbf{y}$ and θ , so the p-value (6) becomes $p_B = \frac{1}{K} \sum_k \mathbf{1}[T_{\chi^2}(\mathbf{y}^{(k)}, \theta^{(k)}) \geq T_{\chi^2}(\mathbf{y}, \theta^{(k)})]$. This catches overall lack of fit but can miss structured discrepancies.

(b) **Targeted statistics** probe specific features:

- Location of a peak: $T(\mathbf{y}) = \arg \max_i y_i$.
- Symmetry: $T(\mathbf{y}) = \|y_{\text{left}} - y_{\text{right}}\|$ for data components that should be symmetric under the model.
- Smoothness: $T(\mathbf{y}) = \sum_i (y_{i+1} - y_i)^2$ (total variation).
- Tail behavior: $T(\mathbf{y}) = \max_i |y_i|$.

(c) **Use multiple statistics.** Check 3–5 different T ’s, each probing a different model assumption. If all give $p_B \in [0.1, 0.9]$, there is no evidence of misspecification. If one gives an extreme p_B , the specific T that flagged it tells you *what* the model is getting wrong.

2.4 Limitations of posterior predictive checks

- (i) **Double use of data.** The data $\mathbf{y}$ is used both to fit the posterior and to compute $T(\mathbf{y})$. This makes the p-value (6) conservative: it is biased toward 0.5 and can fail to detect real misspecification. The severity of this bias depends on the “effective number of parameters” in the model — it is worst when the model is highly flexible.
- (ii) **Cross-validation variants.** To avoid the double-use problem, one can hold out a subset of the data $\mathbf{y}_{\text{test}}$ and use the remaining data $\mathbf{y}_{\text{train}}$ to fit the posterior. Then the predictive check is performed on $\mathbf{y}_{\text{test}}$. This is sharper but more expensive (multiple MCMC runs, or importance-weight resampling from a single run).
- (iii) **No formal “rejection.”** Posterior predictive p-values are not classical p-values; they do not have a uniform null distribution under the model. They are best treated as continuous diagnostics, not binary hypothesis tests.

3 Bayesian Model Comparison via the Evidence

3.1 The Bayes factor

Suppose we have two candidate models $\mathcal{M}_1$ and $\mathcal{M}_2$, each specifying a forward map $\mathcal{G}_i$, a noise model $p_{\eta,i}$, and a prior $\pi_{0,i}$ on the respective parameter spaces Θ_i . The evidence for model $\mathcal{M}_i$ is

$$Z_i := \int_{\Theta_i} L_i(\mathbf{y} \mid \theta) \pi_{0,i}(\theta) d\theta, \quad (8)$$

which is the normalizing constant of the posterior (3) under model $\mathcal{M}_i$. The **Bayes factor** in favor of $\mathcal{M}_1$ over $\mathcal{M}_2$ is

$$B_{12} := \frac{Z_1}{Z_2}. \quad (9)$$

If $B_{12} > 1$, the data favor $\mathcal{M}_1$; if $B_{12} < 1$, they favor $\mathcal{M}_2$. The Jeffreys scale (a rough guideline): $B_{12} > 10$ is “strong” evidence, $B_{12} > 100$ is “decisive.”

Remark 1 (Built-in Occam’s razor). The evidence (8) is *not* the likelihood maximized over θ (that would be the maximum likelihood, which always favors more complex models). It is the likelihood *averaged* over the prior. A model with more parameters spreads its prior mass over a larger volume, diluting the average. So the evidence automatically penalizes unnecessary complexity: a simpler model that fits the data adequately will have higher evidence than a complex model that fits slightly better but wastes prior mass on parameter regions that the data don’t support. This is the Bayesian Occam’s razor.

3.2 Computing the evidence: thermodynamic integration

Standard MCMC produces samples from $\pi(\theta \mid \mathbf{y})$ but does not compute the normalizing constant Z . **Thermodynamic integration** (TI), also called **power posterior sampling**, is a method for estimating $\log Z$ from a sequence of MCMC runs at different “temperatures.”

Define the **power posterior** at inverse temperature $\beta \in [0, 1]$:

$$\pi_\beta(\theta) := \frac{L(\mathbf{y} \mid \theta)^\beta \pi_0(\theta)}{Z(\beta)}, \quad Z(\beta) := \int L(\mathbf{y} \mid \theta)^\beta \pi_0(\theta) d\theta. \quad (10)$$

At $\beta = 0$, $\pi_0(\theta) = \pi_0(\theta)$ (the prior) and $Z(0) = 1$ (assuming the prior is normalized). At $\beta = 1$, $\pi_1(\theta) = \pi(\theta \mid \mathbf{y})$ (the posterior) and $Z(1) = Z$ (the evidence). The key identity is:

Proposition 1 (Thermodynamic identity).

$$\log Z = \int_0^1 \mathbb{E}_{\pi_\beta} [\log L(\mathbf{y} \mid \theta)] d\beta. \quad (11)$$

Proof. Differentiate $\log Z(\beta) = \log \int L^\beta \pi_0 d\theta$ with respect to β :

$$\frac{d}{d\beta} \log Z(\beta) = \frac{\int (\log L) L^\beta \pi_0 d\theta}{\int L^\beta \pi_0 d\theta} = \mathbb{E}_{\pi_\beta} [\log L].$$

Integrate both sides from $\beta = 0$ to $\beta = 1$:

$$\log Z(1) - \log Z(0) = \int_0^1 \mathbb{E}_{\pi_\beta} [\log L] d\beta.$$

Since $Z(0) = 1$ (normalized prior), $\log Z(0) = 0$, giving (11). $\square$

The algorithm:

Step 1. Choose a grid $0 = \beta_0 < \beta_1 < \dots < \beta_B = 1$. A common choice is a geometric spacing $\beta_j = (j/B)^c$ with $c \approx 5$, which clusters points near $\beta = 0$ where the integrand changes most rapidly.

Step 2. For each β_j , run MCMC targeting $\pi_{\beta_j}(\theta)$ (the power posterior at temperature β_j). This is the same MCMC algorithm you already use, but with the log-likelihood multiplied by β_j .

Step 3. From the samples $\{\theta_j^{(k)}\}_{k=1}^K$ at temperature β_j , estimate

$$I_j := \mathbb{E}_{\pi_{\beta_j}} [\log L] \approx \frac{1}{K} \sum_{k=1}^K \log L(\mathbf{y} \mid \theta_j^{(k)}).$$

Step 4. Approximate the integral (11) by the trapezoidal rule:

$$\log \hat{Z} = \sum_{j=0}^{B-1} \frac{\beta_{j+1} - \beta_j}{2} (I_j + I_{j+1}). \quad (12)$$

Cost: B MCMC runs (typically $B = 10$ – 30), each of the same length as the original posterior run. The total cost is roughly B times the cost of a single inference.

Remark 2 (Variance and error control). The trapezoidal approximation (12) has discretization error $O((\Delta\beta)^2)$ (from the trapezoidal rule) plus Monte Carlo error $O(K^{-1/2})$ at each temperature. The geometric spacing concentrates grid points near $\beta = 0$, where the transition from prior to posterior is sharpest. In practice, $B = 20$ with $K = 5000$ samples per rung is usually sufficient.

3.3 Other methods for computing the evidence

3.3.1 Bridge sampling

Given posterior samples $\{\theta_1^{(k)}\}$ and $\{\theta_2^{(k)}\}$ from two models, bridge sampling estimates Z_1/Z_2 directly using an optimal “bridge function.” The estimator (Meng & Wong, 1996) is:

$$\frac{\hat{Z}_1}{\hat{Z}_2} = \frac{\sum_{k=1}^{K_2} h(\theta_2^{(k)}) L_1(\mathbf{y} \mid \theta_2^{(k)}) \pi_{0,1}(\theta_2^{(k)})}{\sum_{k=1}^{K_1} h(\theta_1^{(k)}) L_2(\mathbf{y} \mid \theta_1^{(k)}) \pi_{0,2}(\theta_1^{(k)})}, \quad (13)$$

where h is the optimal bridge function $h(\theta) \propto [K_1\pi_1(\theta) + K_2\pi_2(\theta)]^{-1}$. Since h depends on Z_1/Z_2 (through the posteriors), the estimator is applied iteratively until convergence.

Bridge sampling requires **overlap** between the two posteriors: the samples from model 1 must have nontrivial density under model 2 and vice versa. This works well for nested models (where one model is a special case of the other) but can fail for very different models.

3.3.2 Nested sampling

Nested sampling (Skilling, 2006) transforms the d -dimensional evidence integral into a one-dimensional integral over the prior mass. Define the “prior volume” $X(\lambda) = \mathbb{P}_{\pi_0}[L(\mathbf{y} \mid \theta) > \lambda]$. Then $Z = \int_0^1 L^*(X) dX$ where $L^*(X)$ is the inverse of $X(\lambda)$. The algorithm:

- (a) Maintain N “live points” drawn from the prior, confined to the region where $L > \lambda_{\min}$.
- (b) At each iteration, remove the point with the lowest likelihood, replace it with a new point drawn from π_0 subject to $L > \lambda_{\min}$, and increment $\lambda_{\min}$.
- (c) The discarded points, weighted by the prior volume they represent, give an estimate of Z .

Nested sampling is popular in cosmology (the **dynesty**, **MultiNest**, and **UltraNest** packages). It works well for moderate dimensions ($\lesssim 30$) but the constrained prior sampling becomes expensive in high dimensions.

3.3.3 Savage–Dickey density ratio (for nested models)

If $\mathcal{M}_1$ is $\mathcal{M}_2$ with a subset of parameters ϕ fixed at a specific value ϕ_0 (e.g., $\mathcal{M}_1$: $\phi = 0$; $\mathcal{M}_2$: ϕ free), then the Bayes factor has a remarkably simple form:

$$B_{12} = \frac{\pi(\phi = \phi_0 \mid \mathbf{y}, \mathcal{M}_2)}{\pi_0(\phi = \phi_0 \mid \mathcal{M}_2)}. \quad (14)$$

That is: the ratio of the posterior density to the prior density at the restricted value, both evaluated under the larger model $\mathcal{M}_2$.

In words: if the data pull the posterior away from ϕ_0 , the numerator is small and $B_{12} < 1$ (the data favor the larger model). If the posterior concentrates at ϕ_0 , the numerator is large and $B_{12} > 1$ (the simpler model suffices).

Cost: Nearly free once you have posterior samples from $\mathcal{M}_2$. You just need a kernel density estimate of the marginal posterior of ϕ evaluated at ϕ_0 . The practical challenge is getting a good density estimate, especially if ϕ is high-dimensional.

4 Hierarchical Inference of Hyperparameters

4.1 The idea

Instead of fixing model parameters (e.g., noise level σ , physical constants α) at their nominal values, treat them as unknowns with (weakly informative) priors and infer them jointly with the primary unknown θ . The joint posterior is

$$\pi(\theta, \alpha \mid \mathbf{y}) \propto L(\mathbf{y} \mid \theta, \alpha) \pi_0(\theta \mid \alpha) \pi_{\text{hyper}}(\alpha), \quad (15)$$

where α collects the hyperparameters and $\pi_{\text{hyper}}(\alpha)$ is a hyperprior (typically vague: wide Gaussian, inverse-gamma, etc.).

4.2 How this detects misspecification

The key diagnostic: if the marginal posterior $\pi(\alpha \mid \mathbf{y})$ concentrates far from the nominal (experimentally measured or physically expected) value α^* , the model is compensating for a structural deficiency by distorting α .

More precisely: let α^* be the “correct” value of the hyperparameter (known from an independent measurement or physical theory). Run MCMC on the joint model (15) and examine the marginal posterior $\pi(\alpha \mid \mathbf{y})$.

- If $\alpha^* \in$ the 95% credible interval of $\pi(\alpha \mid \mathbf{y})$: no tension. The model is consistent with the known value.
- If α^* is far in the tail of $\pi(\alpha \mid \mathbf{y})$: the data are forcing α away from its known value. The model needs $\alpha \neq \alpha^*$ to fit the data, which means something else in the model is wrong. This is a specific, interpretable diagnostic: the *direction* in which α is pulled tells you what physics is missing.

4.3 Advantages over model comparison

Hierarchical inference is a *continuous* version of model comparison. Rather than choosing between discrete models ($\mathcal{M}_1$: fix $\alpha = \alpha^*$ vs. $\mathcal{M}_2$: infer α), you let the data tell you *how much* α needs to deviate from α^* . This is often more informative than a binary Bayes factor, because:

- It gives a direction and magnitude of tension, not just a number.
- It does not require computing the evidence Z (standard MCMC suffices).
- It naturally handles multiple hyperparameters simultaneously.

4.4 MCMC implementation: Gibbs-within-Metropolis

The joint posterior (15) has two qualitatively different blocks of unknowns: the primary unknown θ (high- or infinite-dimensional, e.g., a function $\delta(y)$) and the hyperparameters α (low-dimensional, e.g., a handful of scalars). The natural sampling strategy is a **Gibbs alternation** with specialized updates for each block.

4.4.1 Block 1: Update $\theta \mid \alpha, \mathbf{y}$

Conditional on the current hyperparameter values α , the full conditional for θ is

$$\pi(\theta \mid \alpha, \mathbf{y}) \propto L(\mathbf{y} \mid \theta, \alpha) \pi_0(\theta \mid \alpha). \quad (16)$$

This is exactly the posterior of the original inverse problem, but with the likelihood and prior evaluated at the current α . For infinite-dimensional θ (function-valued unknowns), this is where **dimension-robust MCMC algorithms** are essential:

- **pCN** (preconditioned Crank–Nicolson): proposals of the form $\theta' = \sqrt{1 - \beta^2} \theta + \beta \xi$, $\xi \sim \pi_0$, which are reversible with respect to the prior and whose acceptance rate is independent of the discretization dimension.
- **∞ -HMC** (infinite-dimensional Hamiltonian Monte Carlo): Hamiltonian dynamics on the Cameron–Martin space of the prior, giving gradient-informed proposals that are also dimension-robust.
- **mpCN and MESS** (multiproposal variants): generate multiple proposals per step for improved parallel efficiency and mixing in multimodal posteriors.

These serve as the **Metropolis-within-Gibbs** step for the θ block. The key observation is that conditioning on α does not change the structure of this update at all — it only changes the numerical values plugged into the forward map $\mathcal{G}(\theta; \alpha)$ and possibly the prior covariance operator $\mathcal{C}(\alpha)$. Existing MCMC code for the primary inference needs only to accept α as a parameter input rather than a hardcoded constant.

4.4.2 Block 2: Update $\alpha \mid \theta, \mathbf{y}$

Conditional on the current θ , the full conditional for α is low-dimensional:

$$\pi(\alpha \mid \theta, \mathbf{y}) \propto L(\mathbf{y} \mid \theta, \alpha) \pi_0(\theta \mid \alpha) \pi_{\text{hyper}}(\alpha). \quad (17)$$

Since α is typically 3–5 scalars, a simple Metropolis step (random walk, or adaptive Metropolis, or MALA) suffices. In some cases, conjugacy gives a direct Gibbs draw:

Example 1 (Noise variance σ^2). If $\Sigma = \sigma^2 I_d$ and σ^2 has an inverse-gamma prior $\sigma^2 \sim \text{IG}(a_0, b_0)$, then the full conditional (17) is also inverse-gamma:

$$\sigma^2 \mid \theta, \mathbf{y} \sim \text{IG}\left(a_0 + \frac{d}{2}, b_0 + \frac{1}{2} \|\mathbf{y} - \mathcal{G}(\theta)\|^2\right).$$

This is a direct draw — no Metropolis step needed, no forward model evaluation required (the residual $\mathbf{y} - \mathcal{G}(\theta)$ is already available from Block 1).

Example 2 (Physical parameters entering $\mathcal{G}$ nonlinearly). For hyperparameters like Δ , E_z , or α_R that enter the forward map $\mathcal{G}$ through the BdG Hamiltonian, there is no conjugacy. The full conditional (17) requires evaluating $\mathcal{G}(\theta; \alpha')$ at the proposed value α' , which is a full forward model solve. However, since this is a 1D (or at most 3–4D) Metropolis update, even a simple random walk mixes quickly.

4.4.3 The full Gibbs iteration

One iteration of the hierarchical sampler is:

$$\boxed{\begin{array}{ll} 1. & \theta^{(k+1)} \sim \pi(\theta \mid \alpha^{(k)}, \mathbf{y}) \quad (\text{pCN/mpCN/MESS — one or several steps}), \\ 2. & \alpha^{(k+1)} \sim \pi(\alpha \mid \theta^{(k+1)}, \mathbf{y}) \quad (\text{Metropolis or Gibbs for each component}). \end{array}} \quad (18)$$

4.4.4 Cost analysis

Block 1 requires at least one evaluation of $\mathcal{G}(\theta; \alpha^{(k)})$ per step (this is the dominant cost and is the same as in the non-hierarchical sampler).

Block 2 cost depends on where α enters:

- If α enters *only through the likelihood* and not through $\mathcal{G}$ (e.g., the noise variance σ^2 scales the residual), then Block 2 is essentially **free**: no forward model evaluation is needed, just a re-evaluation of the likelihood at the current residual with the new α .
- If α enters *through the Hamiltonian* (e.g., Δ , E_z , α_R), then accepting a proposed α' in Block 2 requires a full forward model evaluation $\mathcal{G}(\theta^{(k+1)}; \alpha')$. This is one additional solve per accepted α -proposal.

Practical trick: perform several θ -updates (Block 1 steps) per α -update (Block 2 step). The θ -updates are already paying for forward model evaluations, and the hyperparameters α typically mix faster (being low-dimensional) so they do not need to be updated at every iteration. A ratio of 5–10 θ -steps per α -step is a reasonable starting point.

Total overhead relative to the non-hierarchical sampler: if α only includes the noise variance (conjugate update), the overhead is negligible. If α includes physical parameters entering $\mathcal{G}$, the overhead is roughly the fraction of Block 2 proposals that are accepted times one additional forward solve per acceptance — typically a 10–30% increase in total cost.

5 Residual Analysis

5.1 The Bayesian residual

Given a point estimate $\hat{\theta}$ (e.g., the posterior mean $\hat{\theta} = \mathbb{E}_\pi[\theta]$ or the MAP estimate $\hat{\theta} = \arg \max_\theta \pi(\theta \mid \mathbf{y})$), define the **residual vector**

$$\mathbf{r}(\hat{\theta}) := \mathbf{y} - \mathcal{G}(\hat{\theta}) \in \mathbb{R}^d. \quad (19)$$

Under the model (1), if $\hat{\theta}$ were exactly equal to the true parameter θ^* , then $\mathbf{r}(\theta^*) = \eta \sim N(0, \Sigma)$.

5.2 What to check

Standard regression diagnostics apply:

- Normality.** Are the standardized residuals $\tilde{r}_i = [\Sigma^{-1/2} \mathbf{r}]_i$ approximately $N(0, 1)$? Check with a QQ-plot or a Shapiro–Wilk test. Departures suggest non-Gaussian noise or model misspecification.
- Homoscedasticity.** Plot $|\tilde{r}_i|$ against the predicted values $[\mathcal{G}(\hat{\theta})]_i$ or against an experimental variable (e.g., bias voltage V). Patterns (e.g., larger residuals at certain voltages) indicate that the noise model Σ is wrong or that the forward map $\mathcal{G}$ systematically misses features at those parameter values.

- (c) **Independence.** If the data have a natural ordering (e.g., indexed by voltage), check for autocorrelation in the residuals. Positive autocorrelation suggests the model is missing smooth structure in the data.
- (d) **Omnibus chi-squared.** Under the model, $\chi^2 := \|\Sigma^{-1/2}\mathbf{r}\|^2 \sim \chi_d^2$ (approximately, for $\hat{\theta}$ close to θ^*). If $\chi^2 \gg d$, the model does not fit the data.

5.3 When residual analysis is and is not rigorous

The honest statement: the distributional claim $\mathbf{r}(\hat{\theta}) \sim N(0, \Sigma)$ is only valid if:

- (i) The model is correctly specified (the true DGP is (1) with $\eta \sim N(0, \Sigma)$).
- (ii) The estimator $\hat{\theta}$ is close to the true θ^* (i.e., the posterior has concentrated near the truth — the Bayesian consistency condition).

If the posterior has not concentrated — for example because the data are insufficient to identify θ^* , or because the parameter space is too high-dimensional — then $\hat{\theta}$ may be far from θ^* and the residuals $\mathbf{r}(\hat{\theta})$ will reflect both the noise η and the estimation error $\mathcal{G}(\theta^*) - \mathcal{G}(\hat{\theta})$. In this case the residuals are not guaranteed to be Gaussian even if the model is correct.

The practical value: despite the lack of formal guarantees, residual analysis remains the fastest and most intuitive diagnostic. The logic is asymmetric:

- If the residuals look terrible (structured patterns, heavy tails, heteroscedasticity), the model is almost certainly wrong.
- If the residuals look clean, the model *might* be right — but a wrong model can produce innocent-looking residuals if the misspecification lies in a direction the data cannot probe.

This is the same situation as in classical regression: residual analysis catches many forms of misspecification but is not omniscient.

Remark 3 (Connection to posterior predictive checks). The chi-squared residual $\chi^2 = \|\Sigma^{-1/2}\mathbf{r}(\hat{\theta})\|^2$ at the point estimate is closely related to the omnibus posterior predictive check with $T = T_{\chi^2}$ from (7). The difference is that the posterior predictive check averages over posterior uncertainty in θ , while the residual check conditions on a single point estimate. The posterior predictive version is more Bayesian but more expensive; the residual version is quick and dirty but can be sharpened by examining the residuals at *multiple* posterior samples rather than just the point estimate.

6 Recommendations for the Majorana Nanowire Problem

6.1 Recall of the problem setup

We observe a conductance map $\mathbf{G} = \{G_{LL}^i, G_{LR}^i, G_{RL}^i, G_{RR}^i\}_{i=1}^n$ from a proximitized semiconductor nanowire, where the data are modeled as

$$\mathbf{G} = \mathcal{G}(\delta^*) + \eta, \quad \eta \sim N(0, \Sigma), \quad (20)$$

with $\delta^* : [0, L] \rightarrow \mathbb{R}$ the true disorder profile and $\mathcal{G}$ the forward map:

$$\mathcal{G} : \delta(\cdot) \xrightarrow{H_{\text{BdG}}(\delta)} \xrightarrow{S(E; \delta)} \xrightarrow{\text{Landauer-Büttiker}} \mathbf{G}(\delta).$$

The BdG Hamiltonian is $H_{\text{BdG}} = \tau_z[-\frac{\hbar^2}{2m^*}\partial_y^2\sigma_0 - \mu(y)\sigma_0 - i\hbar\alpha_R\sigma_x\partial_y + V_g(y)\sigma_0] - E_z\tau_0\sigma_z + \Delta\tau_x\sigma_0$, where $\mu(y) = \mu_0 + \delta(y)$ and the known parameters are $m^*, \alpha_R, \Delta, E_z, V_g, \mu_0$. The posterior is

$$\pi(\delta \mid \mathbf{G}) \propto \exp\left(-\frac{1}{2}\|\Sigma^{-1/2}(\mathbf{G} - \mathcal{G}(\delta))\|^2\right) \pi_0(\delta).$$

6.2 Specific model validation recommendations

6.2.1 Posterior predictive checks

Generate replicated conductance maps $\mathbf{G}^{(k)} = \mathcal{G}(\delta^{(k)}) + \eta^{(k)}$ from posterior samples. Recommended summary statistics:

- (i) **Omnibus fit:** $T_1 = \|\Sigma^{-1/2}(\mathbf{G} - \mathcal{G}(\delta))\|^2$ (chi-squared at each posterior sample).
- (ii) **Left-right asymmetry:** $T_2 = \|G_{LL}(V = 0) - G_{RR}(V = 0)\|$. A true MZM pair produces quantized ZBCPs at both ends; if the model is correct, the posterior predictive distribution of T_2 should be consistent with the observed asymmetry.
- (iii) **Nonlocal conductance magnitude:** $T_3 = \max_V |G_{RL}(V)|$. This probes whether the model captures bulk transport correctly.
- (iv) **ZBCP width:** $T_4 =$ full width at half maximum of $G_{LL}(V)$ near $V = 0$. Sensitive to temperature broadening and tunnel barrier modeling.

Cost: essentially free once the MCMC run is complete, since $\mathcal{G}(\delta^{(k)})$ has already been evaluated during sampling.

6.2.2 Bayesian model comparison

The most natural model comparisons for this problem:

(A) Disorder only vs. disorder + unknown gate potential.

- $\mathcal{M}_1$: Infer $\delta(y)$ with $V_g(y) = V_g^{\text{nominal}}(y)$ fixed at the experimentally specified gate profile.
- $\mathcal{M}_2$: Infer both $\delta(y)$ and $V_g(y)$ jointly.

If $\mathcal{M}_2$ has decisively higher evidence, the nominal gate profile is inadequate and the conductance data contain information about gate potential variations that the disorder alone cannot explain.

(B) Single-band vs. multi-band BdG.

- $\mathcal{M}_1$: The current single-band model.
- $\mathcal{M}_2$: A multi-band extension with additional sub-bands in the nanowire.

More expensive to implement but directly tests the most significant physical simplification in the model.

Recommended method: Thermodynamic integration (§3.2), since MCMC infrastructure is already in place. For the parametric (~ 3 parameter) version of the problem, nested sampling (via *dynesty*) is also practical and gives the evidence as a direct output.

6.2.3 Hierarchical inference

Promote the following nominal parameters to inferred quantities with weakly informative priors centered at their experimental values:

- (i) **Superconducting gap Δ :** if the posterior on Δ is pulled away from the experimentally measured value, this indicates that the proximity effect is not spatially uniform or that the effective gap differs from the bulk value. One could then extend the model to infer a spatially varying $\Delta(y)$.

- (ii) **Zeeman energy E_z** : should be well-determined by the applied magnetic field, but if the effective g -factor varies along the wire, the posterior on E_z will be pulled.
- (iii) **Noise variance σ^2** (or the full Σ): if the posterior on σ^2 is much larger than the experimentally estimated noise, the model is under-fitting the data (the extra variance compensates for systematic model error).
- (iv) **Spin-orbit coupling α_R** : should be determined by the material, but effective values can differ from bulk measurements in confined geometries.

Cost: Modest — enlarges the parameter space by a few dimensions (one per hyperparameter). The MCMC mixing may slow down slightly but the per-sample cost of $\mathcal{G}$ is unchanged.

6.2.4 Residual analysis

At the posterior mean $\hat{\delta} = \mathbb{E}_\pi[\delta]$, compute the standardized residuals $\tilde{r}_i = [\Sigma^{-1/2}(\mathbf{G} - \mathcal{G}(\hat{\delta}))]_i$ and check:

- (i) **QQ-plot** of $\{\tilde{r}_i\}$ against $N(0, 1)$ quantiles.
- (ii) **Residuals vs. bias voltage V** : plot $\tilde{r}_i$ against V_i . Systematic structure (e.g., residuals consistently positive near $V = 0$ or at the gap edge $eV \approx \Delta$) indicates the model misses features at specific energies.
- (iii) **Residuals vs. Zeeman field V_Z** : plot $\tilde{r}_i$ against $V_{Z,j}$. Structure near the topological phase transition ($V_Z \approx \sqrt{\mu_0^2 + \Delta^2}$) would indicate the model struggles precisely where the physics is most interesting.
- (iv) **Residuals by component**: compare the residual distributions for G_{LL} , G_{LR} , G_{RL} , G_{RR} separately. If one component has systematically larger residuals, the model may be misspecifying the physics at one end of the wire or in the bulk.
- (v) **Chi-squared test**: $\chi^2 = \sum_i \tilde{r}_i^2 \approx \chi_d^2$. Report χ^2/d (should be ≈ 1).

Cost: Negligible (one forward model evaluation at the posterior mean).

6.3 Recommended workflow

In order of increasing effort:

1. **Residual analysis** (free, do always): inspect the residuals at the posterior mean for structure. Takes minutes.
2. **Posterior predictive checks** (cheap, do always): generate K replicated datasets from the posterior, check 3–5 summary statistics. Requires no additional forward model evaluations beyond what MCMC already computed.
3. **Hierarchical inference** (moderate cost, do for key parameters): promote Δ , σ^2 , and possibly E_z and α_R to inferred quantities. Run one additional MCMC chain with the expanded parameter space. Examine the marginal posteriors for tension with known values.
4. **Bayesian model comparison** (expensive, do for key model choices): compute the evidence via thermodynamic integration for the disorder-only model vs. the disorder-plus-gate-potential model. Requires ~ 20 additional MCMC runs.